

Highlights

Local skill, unstable portability: marginal diagnostics do not order the cross-region transfer of pre-fire thermal dryness in Mediterranean wildfire regions

Yunus Emre Cogurcu, Emrehan Metin

- Pre-fire thermal predictors add +0.06 to +0.15 AUC within all five fire regions
- The same block adds +0.004 AUC across 20 transfer directions, interval spans zero
- Highest niche-overlap pair fails both ways; lowest transfers both ways (points)
- Of 20 diagnostics, only conditional direction agreement tracks transfer (ρ 0.84)
- Label-blind adaptation compresses transfer toward chance; only target labels help

Local skill, unstable portability: marginal diagnostics do not order the cross-region transfer of pre-fire thermal dryness in Mediterranean wildfire regions

Yunus Emre Cogurcu^{a,*}, Emrehan Metin^a

^a*Çukurova University, Adana, Türkiye*

Abstract

Pre-fire thermal dryness, measured by land surface temperature, its anomalies and dryness indices, separates a fire year from a normal year. Models built on it are rarely tested outside their training region.

Five Mediterranean wildfire regions were analysed on about 500 m cells, with MCD64A1 labels and spatially blocked validation. Six pre-fire thermal predictors were added to a static and near-static terrain, fuel and greenness baseline. Within every region, ROC-AUC rose by +0.06 to +0.15, and the gain survived coarser blocks and an earlier predictor window.

The same predictors were then transferred between regions. Their mean contribution over twenty ordered directions was +0.004, with an interval spanning zero. Removing the two reversing predictors cost -0.081 of within-region skill, supported in every region, and changed mean transfer by +0.014, whose pair-clustered interval spans zero. The debit is measured and the credit is not.

The failure is conditional. Inside one study area, two fires eleven months

*Corresponding author.

Email address: `ycogurcu@cu.edu.tr` (Yunus Emre Cogurcu)

apart reversed the elevation-burning link, with disjoint bootstrap intervals. Season, year and population all differ there, so place alone is what the design holds fixed.

Twenty candidate diagnostics were rank-correlated against observed transfer. Only two had bootstrap intervals excluding zero, both conditional, the stronger being agreement in the sign of each predictor’s association (Spearman $\rho = 0.84$ over eight pairs). No marginal, niche-overlap or regime measure ordered the matrix; at the point estimate the highest-overlap pair failed both ways and the lowest transferred both ways. Label-blind adaptation by standardisation and CORAL pushed transfer towards chance in fourteen of the twenty directions. Niche and regime measures need labels in both regions too, so only the marginal family can run before deployment, and it fails.

Transfer skill therefore has to be measured, not inferred from similarity. What carries the information is conditional, so the price is target labels, not better unsupervised alignment: thirty-two labelled target blocks recovered 85 to 89 % of the ceiling in three of six directions and 30 to 57 % in the rest. *Keywords:* wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [27]. The pressure to anticipate *where* fire will occur has produced a large and still-growing literature on fire susceptibility mapping. It has recently been synthesised from
5 erature on fire susceptibility mapping. It has recently been synthesised from both a methodological and an application perspective [39, 13]. The domi-

nant pattern is well established. Geospatial predictors are assembled over a study region and paired with a historical record of burned area. A supervised classifier is then fitted, most often a random forest [4]. The resulting susceptibility surface is published with a cross-validated AUC in the 0.85 to 0.95 range.

This paper is about a cost that the pattern does not price. Predictors that describe the *state* of a surface in a particular season are more informative about that season than predictors that describe a place. They are also less portable, as we argue and then show. The relationship they carry is reparameterised locally. It exists everywhere, but it points in different directions in different places. The claim advanced here is a trade-off and not a dismissal: **dynamic state predictors buy local skill at the cost of portability.** Two measurements carry that claim. The dynamic block is worth +0.056 to +0.153 ROC-AUC inside every region and contributes nothing distinguishable from zero between them, with an unstable sign. Removing the reversing predictors costs −0.081 of within-region skill with interval support in every region, while changing mean transfer by +0.014, an estimate whose own interval spans zero. Whether the block helps or harms on transfer is a property of the source-target pair, not of the block.

1.1. Portability goes unmeasured, so the trade-off goes unpriced

A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area and season, often from the same fire event, and where folds are drawn at random over cells, spatial autocorrelation inflates the estimate further. The problem is documented across ecological modelling [31, 29] and is addressed

by spatially blocked cross-validation [37, 23]. Blocking makes the within-region estimate honest. It says nothing about whether the fitted relationship holds elsewhere.

35 Because only that side of the ledger is reported, the trade-off is structurally invisible. A predictor block is adopted on the strength of the increment it delivers inside its training footprint, and the portability it consumes is never entered as a debit, even though any regional product built from locally trained models implicitly promises generalisation beyond that
40 footprint. For fire specifically the transfer question is asked far less often than in adjacent fields. Meteorological fire-danger indices are known not to port cleanly between fire environments [30], and the two recent studies that test model transfer systematically, across Mediterranean countries [7] and US counties [15], both report that it largely succeeds between similar
45 regions. Both transfer models whose dominant predictors are *spatially stationary*: terrain, human modification, night-time lights, population density, long-term climatologies. Each describes a place rather than a season, which is the class our thesis expects to travel well. Species distribution modelling has examined transferability far more systematically [43], and there geographic
50 and environmental similarity do *not* reliably predict transfer success [38, 32].

1.2. Pre-fire thermal dryness is the natural test case

The dynamic class used here is pre-fire thermal dryness, and it is the class most plausibly *expected* to transfer. Standard susceptibility predictors are terrain, fuel type, greenness, long-term climatology and proximity to
55 roads and settlements, and most are static or near-static over the timescale at which fire danger varies. Such a set explains poorly why a particular summer

burned and the preceding one did not. What changes between those summers is the state of the surface: fuel dryness, moisture stress, and how anomalously hot the land surface has become relative to its own baseline. Satellite thermal observation gives direct access to part of that state through fuel moisture content [44], and pairing surface temperature with a vegetation index was established as a live-fuel-moisture estimator for fire-danger rating by Chuvieco et al. [6]. The Temperature-Vegetation Dryness Index [33] formalises that feature space into an internally normalised measure which should, in principle, be less exposed to absolute-temperature offsets between regions than raw land surface temperature. That theoretical portability advantage is tested here.

That pre-fire thermal state carries genuine information about subsequent fire is established [17, 19, 18, 8]. What has not been established is whether the predictive *skill* of a classifier trained on it in one fire region survives strict, label-free application to another. The physics linking moisture stress to combustion is universal, so portability should be most expected for this class, which is what makes it diagnostic: a loss here cannot be dismissed as a peculiarity of a locally defined covariate.

1.3. Why the loss is invisible to the diagnostics in use

A deficit on transfer can arise two ways. Under **covariate shift** the marginal distribution of the predictors differs while the predictor-response relationship is preserved, which is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL [35], an approach with an established remote-sensing literature [36]. Under **concept shift** the conditional relationship itself changes: a predictor pos-

itively associated with burning in one region is negatively associated with it in another. No label-free realignment of input distributions can repair a sign reversal, because the information needed to detect it is exactly the information being withheld. The taxonomy is canonical [25].

This distinction is what makes the failure hard to anticipate. The diagnostics the field relies on to decide whether a model may be applied to new ground, above all area-of-applicability measures built on dissimilarity in predictor space [21, 22, 16], are marginal by construction. A target region can sit well inside the training data’s predictor envelope while the relationship between those predictors and burning points the other way. This paper measures both sides on the same pairs.

1.4. Contributions

Three findings carry this paper. Each is stated at the strength its interval supports, and the nearest prior work is named inside the claim rather than omitted.

Contribution 1. The local-skill and portability trade-off, quantified per direction. For every ordered region pair, the change in transfer skill from adding the dynamic pre-fire thermal block to a matched baseline is set against the within-region increment the same block delivers. The block is worth +0.056 to +0.153 ROC-AUC inside every one of five regions, with every bootstrap interval above zero and the result holding in both analysis populations. Across the twenty ordered transfer directions it contributes +0.004 on average, an estimate whose interval spans zero under every resampling unit the design permits, and its sign is unstable: paired deltas run from −0.148 to +0.132, twelve positive and eight negative, and neither the point

estimates nor those signs depend on the blocking scale. Feature removal measures the cost side directly. Dropping the two reversing predictors costs -0.081 of mean within-region AUC, supported in every region, and changes
110 mean transfer by $+0.014$, whose interval spans zero: the debit is measured and the credit is not (Section 4.6b). The within-region half is not itself novel, since comparable results exist for these landscapes [1, 12]; the paired contrast against portability is. Dimarco et al. [7] transfer a predominantly spatially stationary predictor set successfully across a comparable Mediter-
115 ranean design, and this is the complementary half of that picture rather than a contradiction of it.

Contribution 2. The loss is invisible to the diagnostics that could be run before deployment, and visible to one that cannot. Twenty candidate transferability diagnostics from four families are evalu-
120 ated against the observed transfer outcomes under one bootstrap framework. Area-of-applicability-style predictor-space dissimilarity [21, 22, 16], climatic and geographic distance, learned domain separability, niche-overlap statistics and burn-pattern regime distances all fail to order the matrix; the domain classifier is at ceiling, with $\text{AUC} \geq 0.96$ for every pair. Only two diagnos-
125 tics have intervals excluding zero, and both measure conditional direction agreement, whether each predictor’s signed association with burning points the same way in both regions (Spearman $\rho = +0.84$ and $+0.81$ over eight pairs). Signed associations need burned labels on both sides, so this is a mechanism diagnosis rather than a pre-deployment screen, and the marginal
130 family, which is the only one that can be run before deployment, is the one that fails. The same point survives without any ranking: the pair with the

highest burned-niche overlap in the matrix fails in both directions while the pair with the lowest transfers in both, so similarity is not sufficient for transfer. That contradicts an expectation the fire literature carries [7, 15] and
135 agrees with what species distribution modelling has found [38, 32]. Shift decomposition in applied remote sensing is not itself new [11]; what is added here is the fire application and the head-to-head of marginal, niche-overlap and conditional diagnostics on the same pairs.

Contribution 3. The mechanism is a sign reversal, and it survives holding geography fixed. Predictors do not merely weaken across
140 regions, they reverse the direction of their association with burning, which is why a distance in predictor space cannot see the failure. The sharpest reversal is elevation, whose association points opposite ways in Manavgat and in Bejís and Muğla with disjoint bootstrap intervals. The same reversal
145 appears inside a single study area, between two fires eleven months apart on an identical grid, where season and population also differ but place does not (Section 4.8). Two consequences follow and are reported as supporting rather than leading results: label-free alignment by standardisation and covariance alignment [35] does not repair transfer but compresses it towards
150 chance in fourteen of twenty directions, which we believe is the first application of covariance alignment to fire susceptibility; and because the residual gap is conditional, the resource that closes it is target labels, whose price is measurable, thirty-two labelled 5 km blocks recovering 85 to 89 % of the target’s own ceiling in three of six directions tested, 51 to 57 % in two more
155 and 30 % in the sixth.

Alongside these, a leakage-audited, spatially blocked evaluation and trans-

fer protocol is released with code, configuration and frozen outputs, so that most of this result can be re-run rather than taken on trust. Two components are exceptions and are named in the data-and-code availability statement:
160 the reproduction-check code and the few-shot export’s exact commit are not in the released repository at the commit of record. This matters given evidence that wildfire transfer conclusions are sensitive to evaluation design and task formulation [42].

A companion paper treats the observational layer beneath this one, where
165 preprocessing decisions taken before any model is fitted are shown to carry their own uncertainty budget.

Section 2 reviews the relevant literature. Section 3 describes the study regions, the data, and the modelling, cross-validation, transfer and adaptation protocol. Section 4 reports results. Section 5 discusses them, including
170 limitations. Section 6 concludes.

2. Related work

2.1. Fire susceptibility mapping, and what it reports

Fire susceptibility mapping has been synthesised recently from both methodological and application perspectives [39, 13]. The dominant pattern is stable:
175 geospatial predictors are assembled over a study region, paired with a historical burned-area record, and fitted with a supervised classifier, most often a random forest [4], and the resulting surface is published with a cross-validated AUC in the 0.85 to 0.95 range. That figure is a within-region estimate. Comparable within-region results already exist for the landscapes
180 studied here [1, 12].

2.2. *Pre-fire thermal dryness*

Satellite thermal observation gives repeated access to surface state through fuel moisture content [44]. Pairing land surface temperature with a vegetation index was established as a live-fuel-moisture estimator for fire-danger rating by Chuvieco et al. [6], and the Temperature-Vegetation Dryness Index [33] formalises that feature space into an internally normalised measure, which is why it is often expected to travel between regions better than raw temperature. That pre-fire thermal state carries information about subsequent fire is established [17, 19, 18]. Gelabert et al. [8] found dead fine fuel moisture and its anomalies to be the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether a classifier built on this class survives strict, label-free application to a region it never saw.

2.3. *Spatial validation and transferability*

Random cross-validation over spatially autocorrelated cells inflates skill estimates [31, 29], and spatially blocked designs are the standard remedy [37, 23]. The remedy is not uncontested: blocked schemes have been argued to introduce pessimistic bias of their own [40, 24, 5]. This paper uses spatial blocking throughout and reports every result at three block sizes, so that a reader can see which verdicts depend on the choice.

Blocking makes a within-region estimate honest but says nothing about another region. For that, the field’s instrument is the area of applicability and related predictor-space dissimilarity measures [21, 22, 16], which ask whether a target’s predictor values fall inside the training data’s envelope.

Species distribution modelling has examined transferability far more systematically than fire has [43], and two studies there report that geographic and environmental similarity do not reliably predict transfer success [38, 32], and the transferability of satellite-derived empirical models is an active question
210 in fire remote sensing itself [20].

2.4. Dataset shift and label-free adaptation

The canonical taxonomy separates covariate shift, where the predictor distribution moves but the predictor-response relationship holds, from concept shift, where the relationship itself changes [25]. Covariate shift is in
215 principle correctable without target labels, by per-region standardisation or by covariance alignment such as CORAL [35], and domain adaptation has an established remote-sensing literature [36, 28]. Concept shift is not: no realignment of input distributions can repair a reversal in the sign of an association, because detecting it requires the labels being withheld. Shift
220 decomposition in applied remote sensing is not itself new [11]. We are not aware of a prior application of covariance alignment to fire susceptibility, fire occurrence or burned-area prediction.

2.5. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. [30] provide the long-standing reference point that meteorologically derived danger
225 indices do not port cleanly between fire environments. WildfireGenome [15] performs a leave-one-county-out transfer matrix across seven ecologically diverse US counties and reports strong within-county performance with highly variable off-diagonal transfer, ecologically similar pairs transferring well and

230 dissimilar pairs collapsing; its label, however, is a principal-component composite of federal hazard *indicators* rather than observed burned area, and it applies no adaptation and no shift decomposition. Xu et al. [42] argue that wildfire transfer conclusions depend strongly on evaluation design and task formulation, a caution that applies here as much as anywhere, and which
 235 we address by fixing the analysis protocol in a project log before the diagnostics were computed and by reporting every sensitivity axis. That log is not a formal pre-registration and no independent timestamped registration exists; Section 3.11 states what was fixed and when. For infrastructure at this scale, Kondylatos et al. [14] provide Mesogeos, a 1 km Mediterranean
 240 datacube built for data-driven wildfire modelling.

The nearest neighbour, and the contrast the present paper draws.

Dimarco et al. [7] is the closest Mediterranean analogue. They harmonise 500 m predictors across four Mediterranean countries, fit tree ensembles under five-fold spatial cross-validation with an 80/20 hold-out, and test transfer
 245 both leave-one-country-out and as a full 4×4 matrix. Every transfer exceeds AUC 0.80. Transfers between bioclimatically similar countries score higher, and transfers to Morocco are systematically lower, which they attribute to anthropogenic drivers, fire management practice and data reporting rather than to climate alone. They apply no domain adaptation.¹

250 The two studies share their essential design: Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation and an explicit transfer matrix. Theirs covers four countries and twelve

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

ordered pairs, ours five regions and twenty ordered directions. **What differs is the predictor class.** Their model rests on quantities that are attributes
255 of a place: terrain slope, night-time lights, human modification, population density, and temperature, wind and humidity as long-term seasonal climatologies. Every one is spatially stationary. Ours rests on the state of a surface in one season. Their result and ours are therefore not in conflict; read together they suggest that the relationship between domain similarity
260 and transfer success is **predictor-class dependent**, so that models built on stationary attributes track similarity while models built on dynamic state need not. We present that reading as a live disagreement rather than a settled result: Vesik et al. [38] and Rousseau and Betts [32] align with it from species distribution modelling, while Dimarco et al. and WildfireGenome run
265 against it.

Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatology [26]. We follow their Methods, which we take to be authoritative, and note the discrepancy so that the comparison is transparent. Every statement
270 about their work was checked against the full text, read 2026-08-08, not the abstract alone.

3. Data and methods

3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed: Manavgat 2021 and
275 Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangular area of

interest in EPSG:4326, defined from place coverage rather than from a fire perimeter, and not tuned on burned prevalence, gate outcome or any model metric. One AOI choice was label-informed and is stated as such: the North
 280 Evia box was extended after the legacy box was found to carry an atypically high burned prevalence, the extended geometry was then defined from place anchors, and the legacy variant is retained as a sensitivity (Section 4.7a). The AOIs are deliberately not clipped to fire perimeters, so that unburned cells surrounding each fire form the negative class rather than being excluded
 285 by construction. A sixth region, Kozan 2023, is carried as a negative control and is excluded from all modelling by the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation can post-date the first labelled burning. Window lengths are matched within
 290 a region and vary between them with the event: 56 to 60 days for predictors and 35 to 49 days for labels. A four-year baseline of window-symmetric composites precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and are
 295 reproduced in Table 1.

3.2. Burned-area label and the ~ 500 m analysis grid

Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1) retrieved through Google Earth Engine [10], whose omission and commission characteristics [3] bound every model here. The product is exported onto
 300 the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which duplicates each native ~ 500 m observation across the 30 m pixels beneath

it.

The analysis grid is therefore **reconstructed rather than native**: the 30 m grid is partitioned into non-overlapping blocks of $\text{round}(500 / 30) =$
305 17×17 pixels, giving a nominal cell edge of 510 m, and each cell is identified by its integer block indices `row_500m` and `col_500m`. This approximates the MODIS sinusoidal cell rather than reproducing it, being anchored to the Landsat/EPSG:4326 reference grid, and marginal blocks are truncated. The cell is square in degrees but not on the ground, about 510 m north to south
310 and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a MODIS cell's 0.250 km²; every block-size label in this paper is therefore the north-south dimension. The companion paper develops what follows from that.

A cell's representative burn date is the **mode** of the positive sub-pixel
315 day-of-year values in the block, tested against the region's label window. Because the exported raster carries no out-of-window positives, in this dataset a single in-window positive sub-pixel makes the cell burned. The fraction of positive sub-pixels agreeing with the modal date is recorded as `burn_date_pixel_agreement_fraction` but no agreement threshold is imposed. The
320 label never affects a cell's eligibility for modelling: unburned, all-no-data and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned inside a region's own predictor window are removed from its analysis universe rather than counted as unburned, which ran for Muğla, Evia and Montiferru,
325 is recorded as not run for Manavgat and has no recorded status for Bejís. Burning in earlier years is screened for no region in the five-region cohort;

the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.14.

3.3. *Burned-landcover admissibility gate*

330 Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural vegetation? A region is admitted as a wildfire candidate when that fraction reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead.
335 The gate uses ESA WorldCover classes aggregated to the same reconstructed cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

3.4. *Predictor variables*

340 Ten predictors are used, four baseline and six thermal, all summarised per cell over the predictor window. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL band with the water bit deliberately preserved; the coarse-resolution thermal input is MODIS/061/MOD11A1.

- 345
- **NDVI**, the predictor-window median of Landsat surface reflectance, is the baseline’s one time-varying member.
 - **Elevation** and **slope** come from the Copernicus DEM GLO-30.
 - **Land cover** is ESA WorldCover v200 [45], entering as the dominant class code.

- 350 • **Current LST**, the predictor-window median of Landsat surface temperature in °C.
- **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- 355 • **TVDI and TVDI difference**. The Temperature-Vegetation Dryness Index [33] is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted
360 per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. The companion paper reports what that scene dependence does and does not explain.
- **Downscaled LST and fused LST**. A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30
365 m grid; the fused product equals observed Landsat LST wherever that is valid and the downscaled surface only where it is not, so gap-filling never replaces or blends a valid observation. The gap-filled share is 0.11 % to 9.70 % by region. The downscaler’s own inputs include coordinates, which is the one route by which a coordinate-derived surface
370 re-enters a feature set from which Section 3.13 excludes coordinates; the companion paper measures what the increment is worth without these two channels.

3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the ~ 510 m cell, computed from valid
375 30 m pixels only, with the valid fraction recorded. A cell is `valid_for_`
`modeling` when it meets a 30 % valid-pixel floor and no thermal channel
is missing. Two populations are then defined. The **primary** population
is natural vegetation, cells whose combined tree, shrub and grass fraction
reaches 0.50, which excludes cropland from every burnable mask. The **sec-**
380 **ondary** population is all valid cells, reported throughout as a sensitivity.
Every headline result is given for both.

3.6. Feature sets and classifier

Two nested feature sets are compared. The **baseline** set is NDVI, el-
elevation, slope and land cover. The **thermal** set is the baseline plus the
385 six thermal channels. Land cover is one-hot encoded and numeric features
are standardised inside the fitting pipeline. The classifier is a random for-
est [4] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced
class weights and `random_state = 42`, identical for every region, popula-
tion, feature set and transfer direction, so that no comparison in this paper
390 is confounded by a model choice.

3.7. Spatial-block cross-validation and bootstrap uncertainty

Cross-validation is spatially blocked. Blocks are squares of the analysis
grid formed as `row_500m // B` by `col_500m // B`, and folds are drawn over
whole blocks, so no block is split between training and test. Five folds are
395 used. The block size B is reported at 2, 10 and 20 cells, about 1, 5 and 10
km, for the within-region analysis.

Uncertainty is a spatial-block bootstrap: blocks are resampled with replacement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the interval. Because it is blocks that are resampled, what bounds an interval’s reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few such blocks it is stated as indicative rather than as an interval.

3.8. Cross-region transfer protocol

For each ordered pair of regions the model is fitted on the source population and applied to the target population with **no target label used at any point**: no refitting, no threshold selection, no calibration. Both the baseline and the thermal feature set are transferred, so that the thermal block’s contribution is a paired difference within a direction rather than a comparison across directions. All twenty ordered directions among the five regions are computed.

3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.

Region-wise z-score. Each region’s numeric features are standardised using its own statistics, source statistics from source data and target statistics from target data, never pooled. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-order marginal offsets.

CORAL after region-wise z-score. Starting from the z-scored features, the source covariance is aligned to the target covariance by the standard whitening-recolouring map [35], with regularisation $\lambda = 10^{-5}$ and eigen-

values floored at 10^{-12} . Critically **the transform is applied to the source only**; the target is left exactly as it is. The classifier is refitted on the aligned source. Neither variant sees a target label, and both are verified label-blind at run time.

425 λ sensitivity was assessed over nine values from 0 to 10^{-1} on four directions in both feature families, moving transfer AUC by at most 0.014 in any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ lies outside the released sweep; the companion paper reports it.

3.10. *Transfer-gap decomposition and the concept-shift diagnostic*

430 For each direction the gap between the target’s own within-region skill and the raw transfer result is split into a part the best label-free adaptation recovers and a part it does not, with the recovered fraction defined as (adapted – raw) / (within – raw), signed and unclipped, and its interval from the same paired bootstrap. The recovered part bounds what covariate-
435 level correction can achieve; the remainder is the conditional residual.

The mechanism is diagnosed by **signed univariate association**. For each numeric predictor the raw ROC-AUC of that predictor against **burned** is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is read as a direction rather than as weakness. A reversal is
440 called bootstrap-supported only when the two regions’ intervals are disjoint, under the same 10-cell spatial-block bootstrap.

3.11. *Transferability diagnostics versus transfer*

Twenty candidate diagnostics from four families are each rank-correlated against the same target quantity, the raw thermal transfer AUC over the

445 twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

The **marginal** family includes area-of-applicability-style dissimilarity in 450 scaled, importance-weighted predictor space [21, 22, 16], climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D [34] and Warren’s I [41] over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The 455 **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

Two properties of this design are stated in advance because they bound 460 what it can show. The candidate set was fixed in a project analysis log before the correlations were computed; that log is not a formal pre-registration and no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a 465 diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

3.12. Interventions

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and

470 evaluated on the held-out region, with folds blocked as before. This asks whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors whose signed association reverses between regions are dropped and everything is refitted, within-region and across every direction. This measures both sides
475 of the trade-off on the same footing: what the removal costs locally and what it buys on transfer.

3.13. Leakage control

An explicit forbidden-column set is enforced at every model fit. Coordinates (`lon`, `lat`, `row`, `col` and their normalised forms), every burn-date and
480 label-provenance column, and the agreement fraction are excluded from all feature sets, and the enforcement runs as an assertion rather than a convention. The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking prevents a cell from sharing a fold with its own neighbours.

Reproducibility. All randomness uses seed 42 and the bootstrap uses
485 1000 replicates throughout. The transfer and adaptation analysis runs in an environment separate from the upstream pipeline’s, so every within-region model was refitted there and compared against the frozen upstream output, and the independently implemented adaptation against the pipeline’s own;
490 the within-region comparisons agree exactly and the twenty transfer directions to within 1.6×10^{-7} . Because random-forest fits are not bit-identical across library versions, all numbers here were produced under, or verified against, scikit-learn 1.9.0, and cross-region point estimates carry an implementation tolerance of about ± 0.02 to 0.03 if that version is not fixed. The

495 companion paper reports the version sensitivity and the reproduction check
in full.

Sensitivity analyses. Every headline result is repeated across two analysis populations, three spatial-block sizes, the CORAL regularisation sweep and both feature sets. Where a conclusion depends on one of those choices, 500 the dependence is reported rather than resolved by choosing the favourable setting.

3.14. Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla is the one region for which a second fire event is analysed on the identical AOI and analysis grid, which allows the direction of feature-label 505 associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening on it, matching the 2021 durations exactly.

This is not a clean temporal-transfer design and is not presented
510 **as one.** The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is same-geography event-to-event. The pair is deliberately kept out of the twenty-direction matrix, and that exclusion is enforced in released code rather than 515 merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion applies to the 2022 arm alone, masking every cell that burned in 2021 so the previous year’s scar does not enter the following year’s analysis: it removes 3,073 cells, of which 2,941 belong to the primary population, taking it from

520 41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of
38,790 cells and three byte-identical static predictors, and in the 2022-to-2021
direction every target positive lies outside the source training population.
Section 4.8 states what follows for how these two numbers may be read.

4. Results

525 4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate as wildfire candidates, and the negative control fails it as intended. Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and the verdict *cropland-dominated control*, and is excluded from all modelling. The separation is not marginal: the five admitted regions carry natural-vegetation
530 fractions of 0.723 to 0.991, so the threshold falls in an empty interval rather than between neighbouring cases. This establishes that the gate discriminates burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 does not distinguish.
535

North Evia is analysed on an extended AOI. Relative to the legacy box, the extended box leaves the burned scar essentially unchanged while cutting overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to 0.287. The effect of that choice on transfer is reported in Section 4.7a.

540 4.2. Within-region: the thermal increment replicates in five regions

In every region, adding the six thermal predictors to the baseline increases spatially blocked out-of-fold ROC-AUC. The increment’s bootstrap interval

Table 1: Region summary (recap of Table 1; final numbering at assembly). Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, **burnable_tree_shrub_grass and valid_for_modeling == True**, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural- veg frac- tion	Gate verdict
Manavgat 2021	24,150	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejís 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (ex- tended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Monte Fitz 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

excludes zero in all five regions at 1 km and at 5 km blocking, which are the two scales this design supports as intervals. At 10 km the point estimates
545 hold, from +0.048 to +0.154, but they rest on 6 to 33 positive-carrying blocks and are reported as indicative rather than as intervals, for the reason given in the table note.

Table note (resampling units). The bootstrap resamples spatial blocks, so what bounds an interval’s reliability is the number of blocks that carry
550 at least one burned cell. Those counts fall quickly as blocks coarsen. At 2 cells they are 235 (Manavgat), 302 (Bejís), 843 (Muğla), 716 (Evia) and 192 (Montiferru), out of 5 439, 3 967, 11 316, 2 566 and 743 blocks. At 10 cells they are 28, 19, 70, 41 and 16, out of 237, 176, 576, 155 and 35. At 20 cells they are **12, 6, 33, 15 and 6**, out of 60, 48, 167, 50 and 12. No
555 bootstrap replicate was invalid at any block size. An equal-tailed percentile interval built on six positive-carrying blocks has no meaningful coverage, and Montiferru at 20 cells additionally feeds only 12 groups into a 5-fold grouped split, so its models train on about ten blocks each. **The 20-cell row of this table should be read as indicative rather than as an interval.**
560 The 10-cell row, where every region has 16 to 70 positive-carrying blocks, is the coarsest blocking this design supports properly, and the increment holds there in all five regions. Source: `paper/referee2_numbers.md`, block C, counted from the frozen per-cell prediction tables.

This is the first half of the trade-off. It is not itself novel, and it is reported
565 because the second half is measured against it.

Table 2: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.8); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

4.3. Cross-region transfer, and what label-free adaptation does to it

Raw transfer is heterogeneous and includes anti-predictive directions. Raw target AUC spans 0.326 to 0.686. Twelve of twenty directions are above chance with interval support, six are *below* chance with interval support, and two intervals span 0.5. Even the best raw transfer, Evia→Manavgat at 0.686, remains far below that target’s own within-region thermal performance of 0.870; across all directions the raw deficit against the within-region reference is 0.184 to 0.592 AUC. Those counts belong to 1 km blocking; at the more conservative 5 km blocking the same points give nine above, four below and seven uncertain, with no direction changing side of the chance line.

Raw transfer is heterogeneous and includes anti-predictive directions. Raw target AUC spans 0.326 to 0.686. Twelve of 20 directions are above chance with CI support, six are *below* chance with CI support (both directions of Manavgat↔Bejís and Manavgat↔Muğla, plus Bejís→Evia and Evia→Bejís), and two intervals span 0.5. Even the best raw transfer (Evia→Manavgat, 0.686) remains far below the target’s own within-region thermal performance (0.870); across all directions the raw deficit against the within-region reference is 0.184 to 0.592 AUC. Those counts belong to the 2-cell blocking of Table 3. At the more conservative 10-cell (~5 km) blocking the same points give 9 above, 4 below and 7 uncertain, with no direction changing side of the chance line (Section 4.7g, Table R13). Four of the six below-chance directions keep their support there: Manavgat→Bejís, Muğla→Manavgat, Bejís→Evia and Evia→Bejís. Bejís→Manavgat and Manavgat→Muğla lose it and carry no verdict. The qualitative statement is unchanged. The counts should not

Table 3: Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.462]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.513, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.550 [0.524, 0.575]	0.549 [0.525, 0.574]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.582 [0.539, 0.624]

be read as exact.

Label-blind adaptation compresses the matrix toward chance rather than repairing it. Under region-wise z-scoring the twenty directions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half the
595 raw spread, with no adapted direction exceeding 0.63 against within-region references of 0.859 to 0.918. Adaptation raises the failing directions and degrades most of those that already transferred. Taking the better of the two adaptations per direction, 14 of the 20 end closer to chance than they began and 6 end further from it; five of those six involve Montiferru, the smallest and
600 last-added region, and move upward, while the sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to 6 split should be read at the precision of limitation (x) in Section 5.9, since Bejís→Manavgat is counted as compressed on a margin of 0.001.

In the five directions where raw transfer was below chance, the best label-
605 free method recovers at most 34 % of the gap to the within-region reference, so the remaining conditional fraction is at least 0.66 everywhere. Seven directions show *negative* recovery, meaning adaptation moves the score away from the reference; in six of those raw transfer was already above chance and adaptation destroyed that advantage. Label-free alignment therefore does
610 not act as a repair mechanism.

4.4. *Transferability diagnostics: only conditional similarity orders transfer*

Twenty candidate diagnostics from four families were each rank-correlated with the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap.

615 *Table note (power): the effective sample is 10 unordered pairs (6 for the*

Table 4: Transfer-gap decomposition (four-AOI set, 12 directions). Within = target’s within-region thermal AUC; best adapted = the better of z-score/CORAL; recovered fraction = (adapted – raw)/(within – raw), signed and unclipped, with paired bootstrap CI (1000 replicates). Montiferru directions are not part of this decomposition (per-pair absolute decompositions exist without fraction CIs). The status column asks whether the *adapted* value clears chance and uses the 2-cell adapted intervals of Table 3. The adapted arms were not recomputed at the coarser blocking of Section 4.7g, which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.550 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

Table 5: All transferability diagnostics versus raw thermal transfer (20 ordered directions). Spearman ρ with pair-based bootstrap 95% CIs. Exp. = expected sign. Rows with n = 12 exist only for the four-AOI subset. The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y x) conditional	+	16	+0.84 [+0.58, +0.88]	yes
Cosine, supported features	P(y x) conditional	+	16	+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	P(y x) conditional	+	20	+0.50 [−0.17, +0.83]	no
Vector	P(y x) conditional	+	20	+0.27 [−0.36, +0.77]	no
Spearman, all 9	conditional				
Agreement count, all 9	P(y x) conditional	+	20	+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	P(x y=1) niche	+	20	+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	P(x y=1) niche	+	20	+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	P(x y=1) niche	+	20	+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	P(x y=1) niche	+	20	−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	P(x y=1) niche	−	20	−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	P(x) marginal	−	20	−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	P(x) marginal	−	12	−0.10 [−0.54, +0.43]	no
Predictor-space p95 dissimilarity	P(x) marginal	−	12	−0.08 [−0.59, +0.49]	no

12-direction rows; 8 for the supported-conditional rows). The two directions of a pair are not independent and every pair shares regions with three others. Intervals of width ± 0.5 to 0.8 cannot rule out moderate true correlations; null rows are "not shown to order transfer", not "shown not to".

620 **Only two diagnostics have intervals excluding zero, and both are conditional.** The stronger is the sign-agreement fraction over interval-supported features, at $\rho = +0.84$ [$+0.58, +0.88$]; the cosine variant reaches $+0.81$. Every marginal measure fails, including area-of-applicability-style dissimilarity in predictor space, climatic distance and geographic distance, 625 and so do the niche-overlap and regime families. The learned domain classifier is at ceiling, separating source from target at $\text{AUC} \geq 0.96$ for every pair, which makes it useless as an ordering instrument precisely because it always succeeds.

Two limits are stated with the result rather than after it. The conditional 630 index is evaluated at its own design size: its tie structure caps the achievable Spearman at $+0.861$, the observed $+0.840$ therefore sits essentially on that ceiling, and the exact one-sided permutation p is 0.0060 , the smallest this tie structure can produce, against a Bonferroni threshold of 0.0026 over the nineteen computed variants. No outcome of this diagnostic could have cleared 635 family-wise correction on ten effective pairs. And signed associations require burned labels in both regions, so the family that works is not available before deployment while the family that fails is.

4.5. The contrast pair: similarity is not sufficient

The clearest single view of Table 5 needs no ranking at all. Manavgat 640 and Muğla lie in the same country and fire year, roughly 200 km apart, and

their burned cells occupy the most similar environmental envelope of any pair in the matrix, with per-feature Schoener’s D of 0.77 to 0.89. Yet five of nine feature-response directions point opposite ways, including elevation, whose signed AUC is 0.374 [0.289, 0.471] in Manavgat against 0.611 [0.532, 0.690] in Muğla with disjoint intervals, and all four absolute thermal channels. Transfer is below chance in both directions at the point estimate, 0.470 and 0.401. Both directions also sit deep inside the nominal area of applicability, at 0.875 and 0.531 of target cells inside the weighted region.

Bejís and Montiferru sit at the opposite extreme, with burned envelopes that barely overlap and the most dissimilar values on every overlap measure, and they transfer above chance in both directions. That half of the contrast rests on point estimates, since neither direction carries a verdict at 5 km blocking, and the marginal applicability audit was never produced for Montiferru. The claim is one of *sufficiency*: similarity does not guarantee transfer, and dissimilarity does not preclude it. Established by coexisting counterexamples, it does not depend on the number of pairs available.

4.6. Interventions: pooling and feature removal obey the same conservation

(a) Pooled multi-region training. Training on the pooled populations of four regions and testing on the held-out fifth does not rescue transfer. At the point estimate the pooled thermal model never beats the best single-source pairwise transfer for any target, with shortfalls of 0.02 to 0.22, and it remains 0.28 to 0.50 AUC below the within-region ceiling. For two targets it falls below the pairwise mean and below chance, though only Bejís is below chance with interval support, at 0.417 [0.369, 0.467].

(b) Removing the direction-reversing features. Retraining without

the two predictors whose signed association reverses between regions measures both sides of the trade-off on one footing. It costs -0.081 of mean within-region AUC, supported in every region, and changes mean transfer by $+0.014$, an estimate whose pair-clustered interval spans zero. The debit
670 is measured and the credit is not, and no exchange rate between them is asserted. This is the paper’s most direct statement of the trade-off: the predictors that carry local skill are the ones that carry the instability.

4.7. Sensitivity analyses

(a) **Evia AOI and prevalence.** Repeating the raw transfer arms with
675 the legacy, high-prevalence Evia box leaves every qualitative conclusion unchanged: thermal raw transfer AUCs move by at most a few hundredths and no direction changes side of the chance line.

(d) **CORAL regularisation.** Over nine λ values from 0 to 10^{-1} on four directions, CORAL transfer AUC moves by at most 0.014 within any
680 direction and 0.008 within the thermal family, so no CORAL-dependent conclusion here is sensitive to λ in that range. The companion paper reports the canonical $\lambda = 1$, which the released sweep omits.

(g) **Blocking scale.** Recomputing the transfer quantities at 10-cell ($\approx$ 5 km) blocking from the frozen per-cell predictions widens the intervals and
685 moves the verdict counts, from ten positive, seven negative and three uncertain at 1 km to six, four and ten at 5 km. **No direction changes side of the chance line and no point estimate changes sign.** The counts are the fragile part of this paper and the sign pattern is the robust part.

4.8. *The same geography, a second fire: reversal with place held constant*

690 Every result above compares different places. Muğla admits a stricter test, because a second fire occurred inside the identical AOI on the identical analysis grid: the 2021 event, with its predictor window closing on 28 July, and the 2022 event, whose matched window closes on 20 June. Region, bounding box, cell definition, feature registry and processing chain are the same. What the design holds fixed is place; what it does not hold fixed is
695 season, since the 2022 fire ignites about six weeks earlier, nor population, since the 2022 arm is defined by removing the 2021 scar.

Elevation reverses with bootstrap support. In 2021 higher ground burned preferentially, at 0.611 with its interval entirely above 0.5; in 2022
700 lower ground did, at 0.296 with its interval entirely below. The two intervals are disjoint, 0.532 against 0.355, and the difference is -0.317 $[-0.414, -0.220]$. The same predictor, the same region, the same grid, and an association that points the opposite way in two fires eleven months apart. The four absolute thermal channels change side as well, from bootstrap-supported
705 lower-values-burn in 2021 to higher-values-burn point estimates in 2022.

Two structural properties must be read alongside these numbers, because they have no analogue in the twenty-direction matrix. The two arms are not disjoint samples: they share 38,789 of the 2022 arm’s 38,790 cells, and elevation, slope and land cover are identical to the digit across all 73,098
710 grid cells, so only NDVI and the six thermal channels carry new information between them. And the removal is the target’s own positive class, so in the 2022 to 2021 direction not one target positive is present in the source training population while 38,789 of 38,819 target negatives are, and membership of

Table 6: Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against **burned**, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.14). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

the source training set alone separates the 2021 target’s classes at ROC-
715 AUC 0.9996. We therefore cannot bound what that asymmetry does to the
two transfer numbers, only state that a direction with this structure is not
comparable to the twenty between-region directions. The elevation reversal is
the finding; the structural asymmetry is a competing explanation this design
cannot exclude. The known part of the bias runs the safe way: the removed
720 cells are high, with a median of 563 m, and unburned in 2022, so removing
them raises the 2022 elevation AUC and makes the reversal smaller rather
than larger.

4.9. What target labels cost: the recovery curve

Everything above measures a failure; this prices it. If the residual gap
725 is conditional and label-free alignment cannot close it, the missing resource
is target-conditional information, and the direct way to supply it is target
labels. The frozen few-shot diagnostic answers how many for three regions in
all six ordered directions, using one 10-cell (≈ 5 km) spatial block as the unit
of labelling effort, budgets of 0 to 32 blocks drawn under a fixed tier order
730 with seeds independent of any result, and ten repeats per budget. Recovery
is read against a matched target-only ceiling of 0.777 to 0.824 under the same
folds and blocking.

Thirty-two labelled blocks recover **85 to 89 % of the target’s own
ceiling in three of the six directions**, two of which started below chance
735 and the third at 0.583; 51 to 57 % in two more, which are the directions
where the conditional gap is widest; and 30 % in the sixth. That budget
is 7 to 20 % of the target’s natural-vegetation population, so it is a real
answer to "what do we do" and not a cheap one. At small budgets the same

intervention damages the direction that already transfers best without any
740 labels. Six directions in three regions cannot support a general label budget
and none is offered.

5. Discussion

5.1. *Principal findings*

Three results carry this paper. First, the pre-fire thermal block raises
745 spatially blocked within-region ROC-AUC by +0.056 to +0.153 in every one
of five Mediterranean regions, with bootstrap support at 1 km and 5 km
blocking, and contributes +0.004 on average across twenty ordered transfer
directions with an interval spanning zero and an unstable sign. Removing
the two reversing predictors costs -0.081 of within-region skill, supported in
750 every region, and buys +0.014 of transfer whose interval spans zero. Sec-
ond, of twenty candidate transferability diagnostics only two have intervals
excluding zero and both are conditional, while every marginal measure fails,
including the area-of-applicability family that is the only one runnable before
deployment. Third, the mechanism is a reversal in the sign of the predictor-
755 burning association, and it persists inside one study area across two fires.

5.2. *Why the thermal increment is real but local*

The within-region increment is not an artefact to be explained away.
It replicates in five independent regions, survives coarsening of the spatial
blocks to ~ 5 km with its interval intact, persists in the secondary all-valid
760 population, and strengthens rather than weakens when the predictor win-
dow is closed earlier. The transfer failure is therefore not evidence that the
thermal signal is spurious. It is evidence that the fitted relationship is *local*.

The natural objection is that different Mediterranean regions are simply different systems, with distinct fuels, terrain and fire histories, so that
765 a predictor meaning one thing in one place and another elsewhere is not instability but a comparison of two different systems. The two Muğla events speak directly to that. Region, AOI, analysis grid, feature registry and processing chain are identical, and the static predictors are identical cell by cell, yet elevation’s association with burning reverses with disjoint intervals,
770 0.611 [0.532, 0.690] in 2021 against 0.296 [0.230, 0.355] in 2022, a difference of -0.317 [-0.414 , -0.220]. Holding geography fixed does not stabilise the direction of the relationship.

The mechanism there is physical rather than statistical. The 2021 season burned as a dispersed complex across the region’s full relief; the 2022 event
775 was a single compact scar confined to lower ground. Two caveats attach and neither is dismissed. Season and year are confounded, because the 2022 event ignites about six weeks earlier, and **that confound cannot be resolved in this study area**: the two events sit 42 days apart in median burn day-of-year, neither year contains a second event at the other’s phase, and a
780 calendar-matched arm would carry nine burned cells against a gate minimum of thirty. Separating year from seasonal phase needs a region with two events at a matching phase in different years, and no region here has one. The design also does not hold the population fixed, since the 2022 arm is the 2021 arm with the 2021 scar removed, which Section 4.8 states in full. What survives
785 both caveats is the narrower reading that *something* reverses the elevation-burning association with geography, grid and pipeline held fixed.

The sharpest supported reversal belongs to elevation, a static predictor,

so instability is a property of the predictor-to-burning mapping generally rather than of thermal channels specifically. The thermal block is where the
790 trade-off is *costly*, not where instability is worst.

5.3. *Why the similarity-based diagnostics fail*

A diagnostic built on distance in predictor space asks whether the target's predictor values look like the training data's. That question is orthogonal to the one that matters when the failure is conditional. A target region can sit
795 deep inside the training envelope while the relationship between those predictors and burning points the other way, and Manavgat to Muğla is exactly that case: 0.875 of target cells inside the weighted area of applicability, and transfer below chance.

The same holds for the niche-overlap and regime families. The pair with
800 the highest burned-niche overlap in the matrix fails in both directions while the pair with the lowest transfers in both, so no monotone function of overlap can order the outcomes. The domain classifier is at ceiling for every pair, which is informative in itself: the regions are trivially separable in predictor space whether or not their transfer works, so separability carries no ordering
805 information here.

5.4. *What the conditional diagnostic is, and what it is not*

The sign-agreement index reaches $\rho = +0.84$ [+0.58, +0.88] against observed transfer, and it is reported with three limits attached rather than after the fact. Its tie structure caps the achievable Spearman at +0.861, so the
810 observed value sits essentially on its own ceiling and the ranking it supports is coarse. Its exact one-sided permutation p is 0.0060, the smallest that tie

structure can produce, against a Bonferroni threshold of 0.0026 over nineteen computed variants, so no outcome could have cleared family-wise correction on ten effective pairs. And it needs burned labels in both regions, which
815 makes it a mechanism diagnosis rather than a pre-deployment screen.

That last limit is the substantive one, and it is why the paper’s practical conclusion is not "use this index instead". It is that the family of diagnostics that can be run before deployment is the family that fails here, and the information that would order transfer is the information a practitioner does
820 not have when the decision is made. The honest statement is that these twenty candidates were **not shown to order transfer**, not that they are shown incapable of it.

5.5. Interventions obey a conservation pattern

Both interventions show the same shape. Pooling four regions never beats
825 the best single-source transfer for any target and stays 0.28 to 0.50 below the within-region ceiling, so aggregation does not manufacture the missing conditional information. Removing the reversing predictors buys a little transfer and pays for it locally, at about six times the cost in within-region skill at the point estimates, though the transfer side’s interval spans zero so
830 no exchange rate is claimed. What transfer gains, the within-region model pays for.

5.6. The regime hypothesis, reported as it happened

A regime-structure explanation was stated in advance and the data confirmed the null. The regime-distance correlation has the wrong sign at the
835 point estimate, the most regime-similar pair fails in both directions, and the

most regime-different pair transfers above chance. One mundane explanation can be set aside: Muğla has by far the largest population, but cutting it to Manavgat’s cell count, and separately to Manavgat’s positive count as well, leaves its transfer behaviour inside the subsampling range in both roles. The error was in the hypothesised grouping, not in the data, and with ten pairs this cannot refute regime typology [2] in general.

5.7. *The empirical contrast with Dimarco et al.*

Dimarco et al. [7] transfer successfully across a comparable Mediterranean design and we do not, and the two results are not in conflict. Their predictors are attributes of a place, ours describe the state of a surface in one season. Read together they suggest that the relationship between domain similarity and transfer success is predictor-class dependent. We present that as a live disagreement, since species distribution modelling aligns with it and the fire literature does not, and it is a reading our data support without proving.

5.8. *Implications*

For practice, transfer skill has to be *measured* rather than inferred from similarity, and the pre-deployment diagnostics currently used for that inference did not order it here. Where a model must be moved, the resource that closes the gap is target labels: thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s own ceiling in three of six directions, and 30 to 57 % in the rest. That is a real answer and not a cheap one, being 7 to 20 % of the target’s natural-vegetation population, and at small budgets the same intervention damages the direction that already transfers best.

For method development, the results point away from better unsuper-
vised alignment. Two label-free methods applied carefully moved fourteen of
860 twenty directions towards chance, and every direction they improved involves
the smallest region. A sign reversal is not a distribution mismatch, and no
realignment of inputs can repair it.

5.9. Limitations

865 The transfer failure is a finding, not a limitation. The limitations are the
boundaries on how far it generalises.

(i) **No meteorological covariates** enter the models, so we cannot say
how the trade-off behaves for a mixed thermal-plus-weather predictor set.

(ii) **Temporal transfer is measured for one region only**, and even
870 there year and seasonal phase are confounded, a confound that cannot be
resolved in this study area for the reason given in Section 5.2. Its 331 burned
cells also leave the thermal reversals unresolved at interval level, and the
pair holds place fixed but not population. Those two arms are addition-
ally the only transfer directions here computed by us rather than read from
875 the pipeline author’s frozen export, with his unmodified code and the same
pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1
[9], whose omission and commission characteristics [3] bound every model
evaluated here. No second burned-area product covers 2021 and 2022 at this
880 resolution; the companion paper reports what an independent active-fire ob-
servation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even
after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072

in three of the others.

885 (v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires multi-year labels.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported
890 numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of ten region pairs**. Both the successes and the failures of Section 4.4 should be read at that power.

895 (viii) **Manavgat’s atypical transfer behaviour remains unexplained**.

It is where the conditional diagnosis bites hardest and where feature removal recovers most. The explanation we could test, that its predictor window was meteorologically extreme, is not supported; the companion paper reports a second candidate, a quality-screening difference in its coarse thermal input
900 whose induced change correlates with elevation. With one fire season per region the remaining candidates are not separable in this design.

(ix) **The interval-support counts are less stable than the point estimates behind them**. Several verdicts sit within a thousandth of their reference value, and at 1 km blocking the published split of ten positive,
905 seven negative and three uncertain turns on a lower bound of -0.00045 . The point estimates and the sign pattern are stable; the counts are not. Every sentence in this paper that leans on an exact count of supported directions should be read at that precision.

6. Conclusions

910 Pre-fire thermal dryness adds a real and repeatable increment to burned-area discrimination. Across five Mediterranean regions, ROC-AUC was raised by +0.06 to +0.15 over a static and near-static terrain, fuel and greenness baseline. That baseline's only time-varying member is the vegetation-index composite. The gain survived spatial blocking at about 5 km, the coarsest
915 scale this design supports as an interval. It also survived a predictor window closed up to two weeks before the first labelled burning.

That skill is local. Paired per direction, the same predictor block adds nothing distinguishable from zero to cross-region transfer. The mean is +0.004 over twenty directions and its interval spans zero under every re-
920 sampling unit the design permits. Its contribution changes sign from one direction to another. The paired deltas run from -0.148 to $+0.132$, twelve positive and eight negative, at either blocking scale. At the conservative 5 km blocking, five to six directions are helped and three to four are harmed with interval support, with the rest carrying no verdict. The block is also
925 the swing factor at the chance line. Removing the two reversing predictors costs -0.081 of mean within-region skill, with interval support in every region, and changes mean transfer by $+0.014$, an estimate whose interval spans zero. The cost of the trade-off is measured; the return is not.

The failure is conditional. The direction of the link between dryness
930 and burning changes from one region to another. None of the similarity diagnostics tested here ordered the transfer matrix: predictor-space distance, domain separability, niche overlap and regime structure all failed. On ten effective pairs those null results mean not shown to order transfer, rather

than shown not to. It is also why label-blind adaptation compresses transfer
935 towards chance instead of repairing it, and why pooled multi-region training
does not escape it.

The practical implication is a change in what is checked before a dynamic-
state fire model is transferred. The question is not whether the target region
lies inside the source’s environmental envelope. In our matrix, the pair with
940 the highest envelope overlap failed in both directions, and the pair with the
lowest overlap transferred in both. Both statements are made at the point
estimate. The question is whether the signed feature-response directions
agree.

That check requires a labelled probe in the target region, and so, on
945 inspection, do the niche and regime diagnostics. Only the marginal family
can be computed before any target label exists, and it is the family that
failed. How large the probe has to be is not established here, so no label
budget is claimed. A signed association with a usable interval may need
fewer labels than a refitted model, but that was not tested. What is clear is
950 that the supervised recalibration that label-free alignment cannot deliver is
not cheap. Section 4.9 prices it for three regions: thirty-two labelled 5 km
blocks recovered 85 to 89 % of the target ceiling in three of six directions, 51
to 57 % in two more and 30 % in the sixth, and at its top budget the labelled
set already holds most of one target region’s burned cells. Where no target
955 labels exist, the transfer performance of such models should be treated as
unknown rather than inferred from similarity.

For model builders, the trade-off should be priced openly. A predictor
block that buys large within-region skill can carry a transfer cost of the oppo-

site sign. Reporting only within-region validation hides that cost completely.
960 Three extensions are left for future work: temporal transfer, meteorological
covariates, and physically normalised dryness variables. Each is a step
towards the self-calibrating thermal monitoring system that motivated this
study.

CRedit authorship contribution statement

965 **Yunus Emre Cogurcu:** Conceptualization, Methodology, Formal analysis,
Investigation, Writing – original draft, Writing – review and editing,
Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Val-
idation, Writing – review and editing.

Declaration of competing interest

970 The authors declare that they have no known competing financial inter-
ests or personal relationships that could have appeared to influence the work
reported in this paper.

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Data availability

All satellite inputs are public and are obtained through Google Earth
Engine. The processing pipeline, its configuration and the frozen numeric
outputs are publicly available; the repository, the commit of record and the

980 licence are given in the data and code availability statement of the Methods,
together with the three components that fall outside that release. No digital
object identifier is minted and no archival deposit exists.

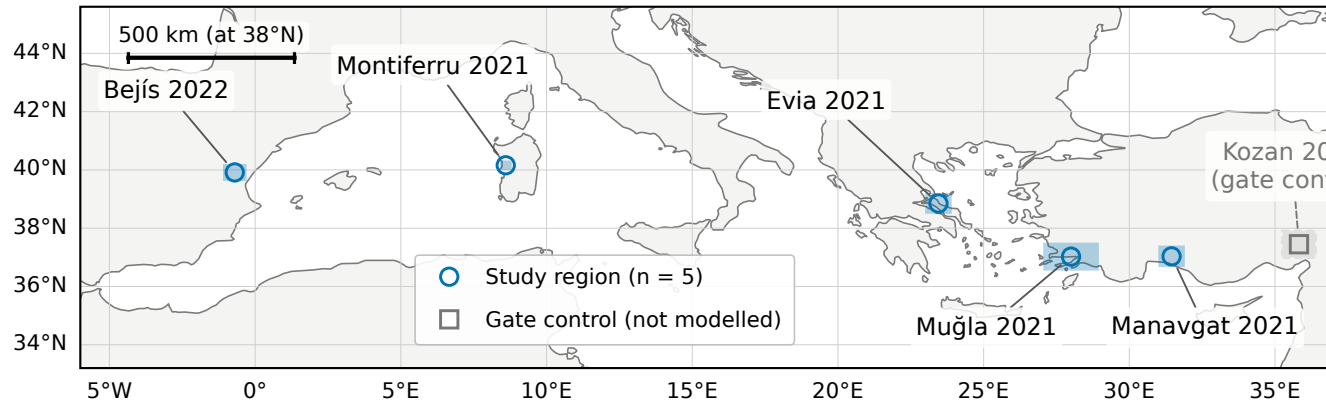


Figure 1: **Study regions and the gate negative control.** Locations of the five wildfire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

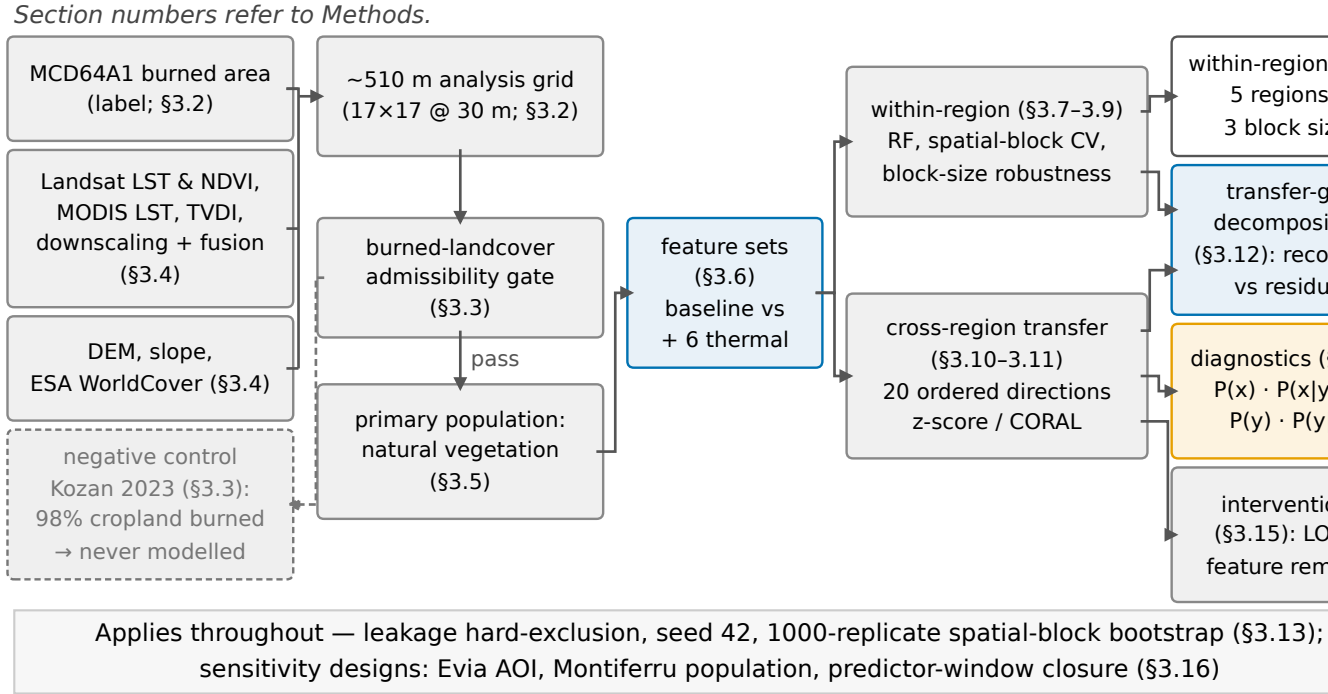


Figure 2: **Processing pipeline and evaluation programme.** Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

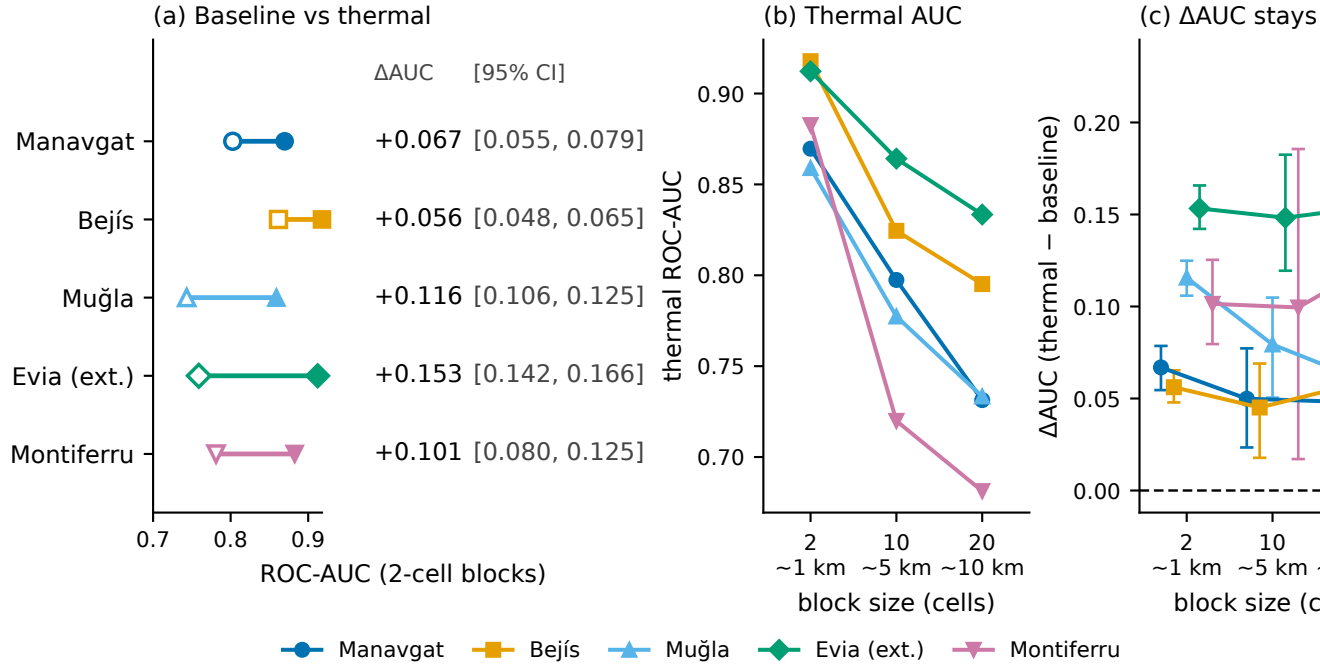


Figure 3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, ≈ 1 , 5 and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region–block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 2, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

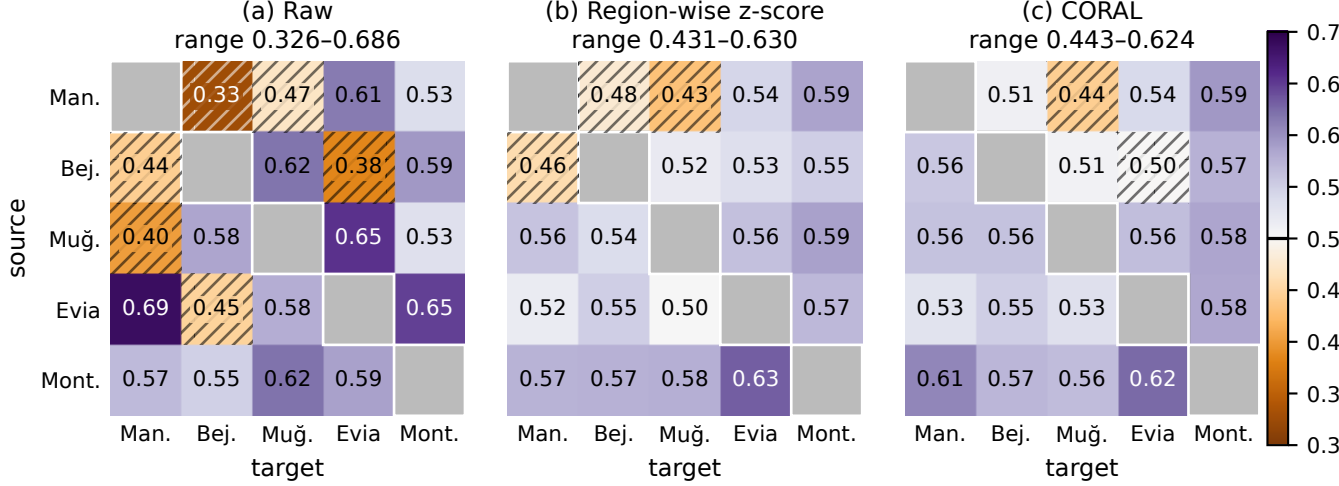


Figure 4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

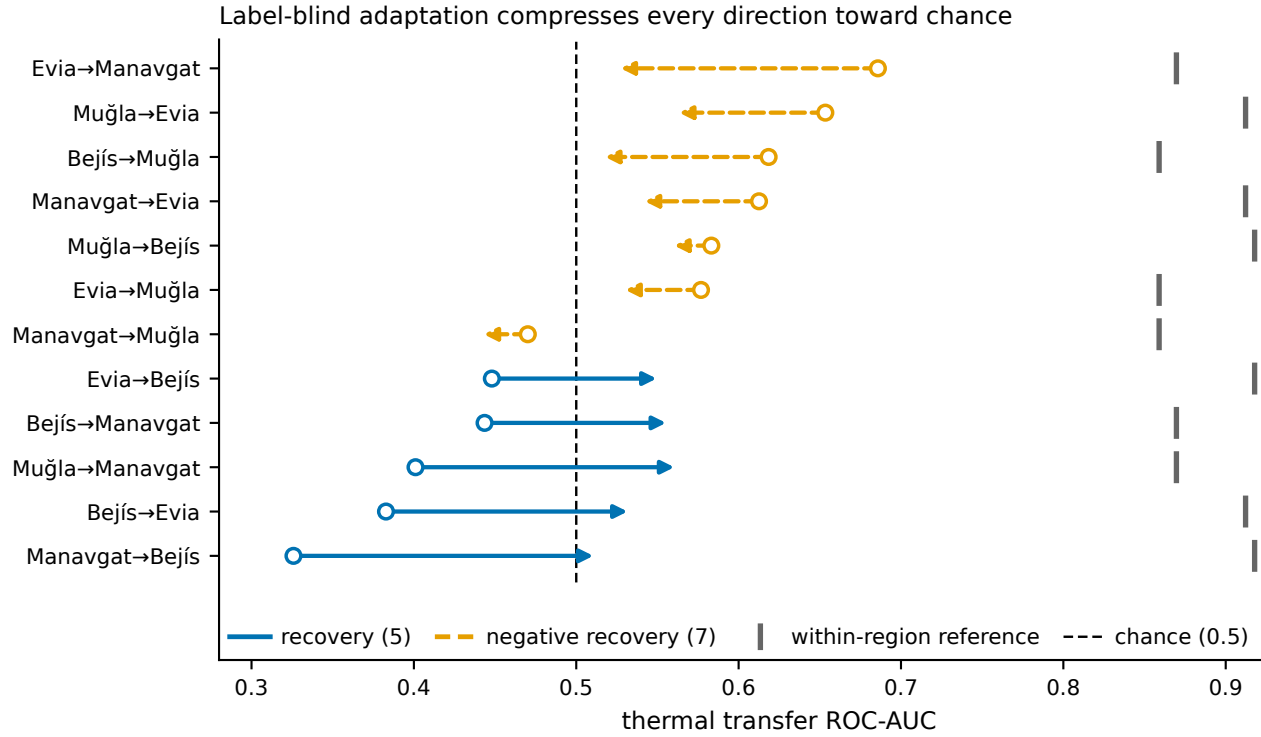


Figure 5: **Label-blind adaptation compresses transfer toward chance rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

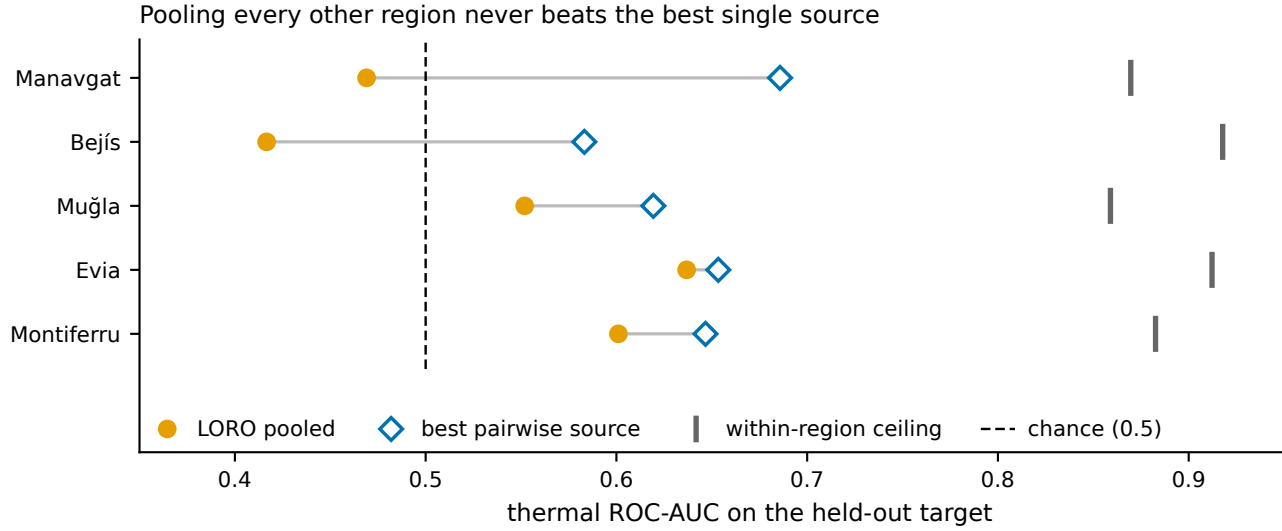


Figure 6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

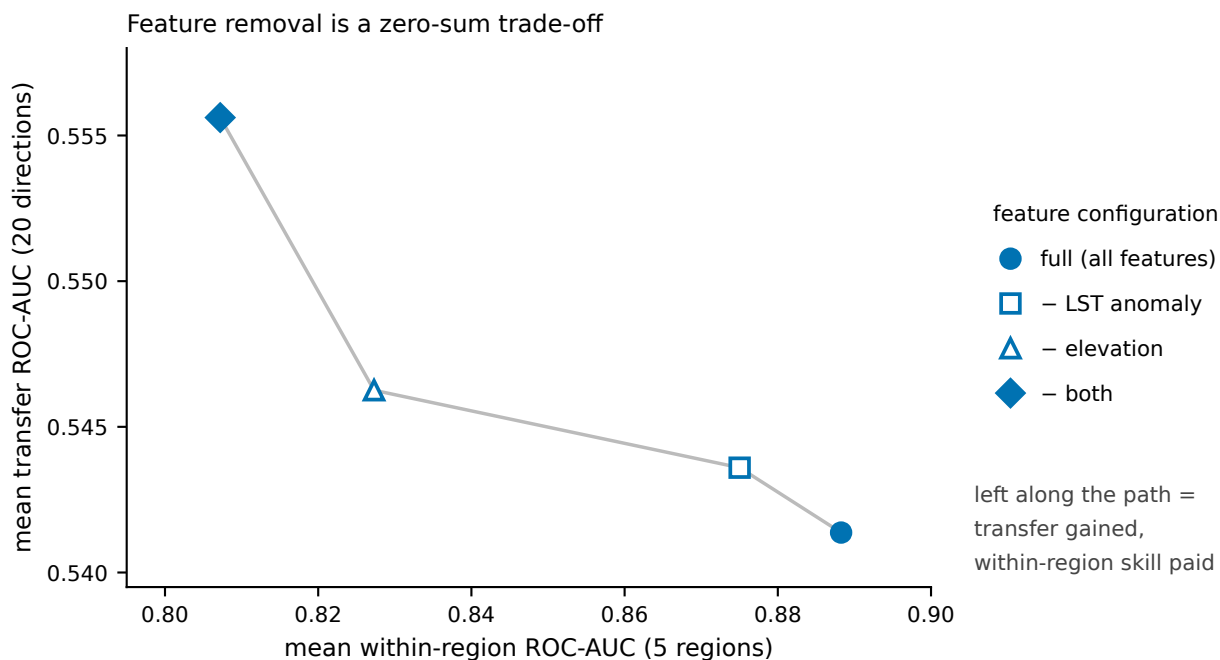


Figure 7: **Removing the direction-reversing features is a zero-sum trade-off.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically (0.541 $\rightarrow$ 0.556) while within-region skill falls monotonically (0.888 $\rightarrow$ 0.807). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The transfer gain is 0.014 AUC and the within-region cost 0.081 AUC, about six times larger: what is removed is locally informative and non-portable, and removing it buys portability by giving up skill rather than by finding transferable signal.

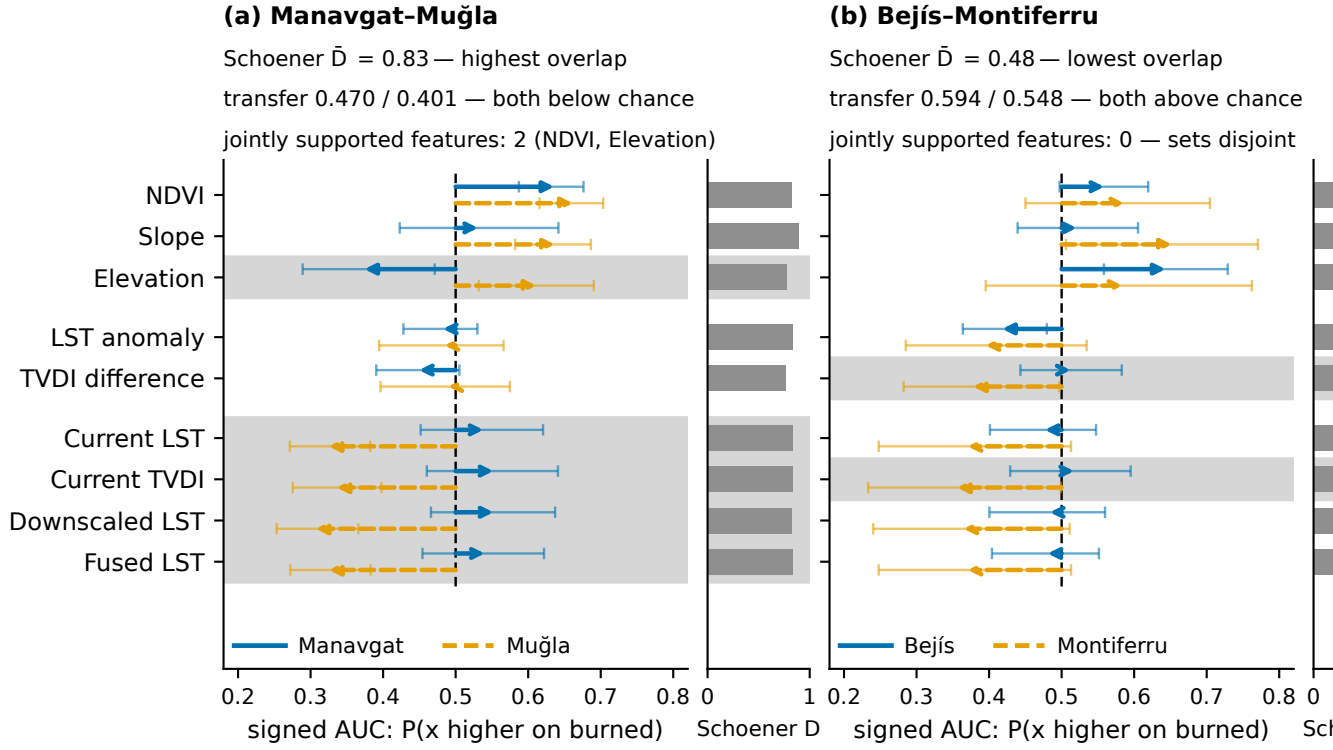


Figure 8: **Burned-niche overlap does not determine transfer: the contrast pair.**

Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat-Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís-Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

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